Table 1
Descriptive statistics of the model variables for rust

Variable	Mean	Median	SD	Confidence Interval
ndev	348.79	304.00	270.09	(346.85, 350.74)
ns	1.03	1.00	0.21	(1.03, 1.04)
nf	6.43	2.00	92.97	(5.77, 7.11)
rexp	74.79	1.13	1321.50	(65.27, 84.32)
sexp	3835.32	673.5	6940.66	(3785.27, 3885.37)
entropy	0.8812	0.2422	1.2184	(0.87, 0.89)
days_to_first_fix	3.46	0.00	47.83	(3.12, 3.81)
exp	4255.27	889.25	7357.31	(4202.22, 4308.33)

Table 2
Descriptive statistics of the model variables for Tensorflow

Variable	Mean	Median	SD	Confidence Interval
ndev	268.72	285.00	140.67	(267.65, 269.81)
ns	1.07	1.00	0.36	(1.07, 1.08)
nf	5.23	2.00	35.58	(4.96, 5.51)
rexp	17.61	1.13	246.65	(15.72, 19.51)
sexp	20180.28	596.5	34515.36	(19915.01, 20445.56)
entropy	0.8850	0.6665	1.0911	(0.88, 0.89)
days_to_first_fix	24.60	0.00	87.73	(23.93, 25.28)
exp	21150.18	780.75	35330.65	(20878.64, 21421.73)

A. Dataset Description

A heat map is a kind of data visualization where different data intensities within a dataset are represented by color gradients. It is frequently used to find correlations, patterns, and trends in data, which makes it easier to quickly display complex information. It conveys potential relationships between the variables shown on the x- and y-axes by using different colors, or varying shades of the same color, to indicate various values. Darker hues or shades typically indicate larger or more quantities of the values shown in the heat map. The reddish color represents the positive correlation, and the bluish color represents the negative correlation among the variables.The distribution shapes, variability, and central tendencies will also be ascertained using descriptive statistics. It also enable us to identify outliers and unusual values. In the below sections, we have visualized each software projects using correlation metrics and descriptive statistics.

A.1. Rust

The heat map of the correlation coefficient among the variables of our interest for the Rust data set is shown in Fig. 1.

Table 1 shows the statistics that can help to understand the distribution and variability of the different variables in the Rust dataset.

A.2. TensorFlow

The heat map of the correlation coefficient among the variables of our interest for the Tensorflow data set is shown in Fig. 2.

Table 2 shows the statistics that can help to understand the distribution and variability of the different variables in the Tensorflow dataset.

A.3. Rails

The heat map of the correlation coefficient among the variables of our interest for the Rails data set is shown in Fig. 3.

Table 3 shows the statistics that can help to understand the distribution and variability of the different variables in the Rails dataset.

Table 3
Descriptive statistics of the model variables for Rails

Variable	Mean	Median	SD	Confidence Interval
ndev	252.61	232.00	169.90	(251.23, 254.00)
ns	1.15	1.00	0.66	(1.15, 1.16)
nf	2.76	1.00	8.20	(2.70, 2.84)
rexp	65.09	1.119	549.86	(60.62, 69.57)
sexp	428.39	77.0	746.29	(422.32, 434.48)
entropy	0.6316	0.00	0.8885	(0.63, 0.64)
days_to_first_fix	106.44	0.00	352.34	(103.57, 109.31)
exp	1713.74	450.0	2613.08	(1692.47, 1735.02)

Table 4
Descriptive statistics of the model variables for VSCode

Variable	Mean	Median	SD	Confidence Interval
ndev	60.60	63.00	27.40	(60.37, 60.84)
ns	1.05	1.00	0.26	(1.05, 1.05)
nf	4.59	1.00	55.96	(4.11, 5.07)
rexp	52.53	1.40	422.99	(48.89, 56.17)
sexp	4739.20	2312.5	5942.17	(4688.06, 4790.36)
entropy	0.7228	0.00	1.0209	(0.71, 0.73)
days_to_first_fix	2.67	0.00	32.59	(2.40, 2.96)
exp	6601.52	4540.75	6804.46	(6542.96, 6660.10)

Table 5
Descriptive statistics of the model variables for WP-Calypso

Variable	Mean	Median	SD	Confidence Interval
ndev	138.94	146.00	75.89	(138.11, 139.79)
ns	1.11	1.00	0.39	(1.11, 1.12)
nf	3.85	2.00	27.52	(3.55, 4.16)
rexp	48.91	1.11	337.92	(45.16, 52.66)
sexp	507.74	206.0	778.86	(499.11, 516.39)
entropy	0.77	0.4262	0.99	(0.76, 0.79)
days_to_first_fix	45.27	0.00	154.18	(43.56, 46.98)
exp	701.74	337.0	998.26	(690.67, 712.83)

A.4. VSCode

The heat map of the correlation coefficient among the variables of our interest for the VSCode data set is shown in Fig. 4.

Table 4 shows the statistics that can help to understand the distribution and variability of the different variables in the VSCode dataset.

A.5. WP-Calypso

The heat map of the correlation coefficient among the variables of our interest for the WP-Calypso data set is shown in Fig. 5.

Table 5 shows the statistics that can help to understand the distribution and variability of the different variables in the WP-Calypso dataset.

A.6. Camel

The heat map of the correlation coefficient among the variables of our interest for the Camel data set is shown in Fig. 6.

Table 6 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table 6
Descriptive statistics of the model variables for Camel

Variable	Mean	Median	SD	Confidence Interval
ndev	47.31	31.00	44.80	(46.81, 47.82)
ns	1.28	1.00	0.83	(1.27, 1.28)
nf	6.77	2.00	41.32	(6.30, 7.24)
rexp	63.45	1.06	682.88	(55.75, 71.15)
sexp	5149.49	844.00	8191.84	(5057.15, 5241.85)
entropy	1.0853	0.8136	1.3254	(1.07, 1.10)
days_to_first_fix	48.74	0.00	225.34	(46.20, 51.28)
exp	15312.83	5702.50	19512.00	(15092.86, 15532.80)

Table 7
Descriptive statistics of the model variables for CoreFX

Variable	Mean	Median	SD	Confidence Interval
ndev	193.68	188.00	109.81	(192.37, 195.01)
ns	1.13	1.00	0.41	(1.13, 1.14)
nf	11.71	2.00	78.34	(10.77, 12.66)
rexp	25.11	1.07	228.37	(22.37, 27.85)
sexp	3538.41	440.00	8407.69	(3437.42, 3639.41)
entropy	1.17	0.83	1.52	(1.16, 1.19)
days_to_first_fix	13.53	0.00	85.95	((12.50, 14.57)
exp	4613.17	788.00	11077.52	(4480.12, 4746.24)

Table 8
Descriptive statistics of the model variables for Django

Variable	Mean	Median	SD	Confidence Interval
ndev	77.25	35.00	81.19	(76.27, 78.23)
ns	1.45	1.00	0.72	(1.45, 1.46)
nf	4.37	2.00	33.82	(3.97, 4.79)
rexp	41.75	1.03	414.70	(36.74, 46.76)
sexp	802.20	268.00	1432.75	(784.91, 819.50)
entropy	0.7102	0.00	1.0183	(0.70, 0.72)
days_to_first_fix	153.80	0.00	377.60	(149.25, 158.36)
exp	2335.92	836.25	3360.53	(2295.35, 2376.49)

A.7. CoreFX

The heat map of the correlation coefficients among the variables of interest in the CoreFX dataset is shown in Fig. 7.

Table 7 shows the statistics that can help to understand the distribution and variability of the different variables in the CoreFX dataset.

A.8. Django

The heat map of the correlation coefficients among the variables of interest in the Django dataset is shown in Fig. 8.

Table 8 shows the statistics that can help to understand the distribution and variability of the different variables in the Django dataset.

A.9. Nova

The heat map of the correlation coefficients among the variables of interest in the Nova dataset is shown in Fig. 9.

Table 9 shows the statistics that can help to understand the distribution and variability of the different variables in the Nova dataset.

Table 9
Descriptive statistics of the model variables for Nova

Variable	Mean	Median	SD	Confidence Interval
ndev	168.82	149.00	118.28	(167.40, 170.25)
ns	1.14	1.00	0.44	(1.14, 1.15)
nf	4.49	2.00	36.56	(4.05, 4.94)
rexp	124.65	1.04	2619.02	(93.01, 156.30)
sexp	363.62	90.0	744.30	(354.63, 372.62)
entropy	0.8764	0.7219	1.0609	(0.86, 0.89)
days_to_first_fix	92.37	0.00	220.29	(89.71, 95.04)
exp	532.47	164.5	980.44	(520.62, 544.32)

Table 10
Descriptive statistics of the model variables for JGroup

Variable	Mean	Median	SD	Confidence Interval
ndev	10.83	11.00	7.07	(10.73, 10.94)
ns	1.17	1.00	0.48	(1.17, 1.18)
nf	2.51	1.00	13.09	(2.32, 2.70)
rexp	47.87	1.0039	1232.16	(29.88, 65.88)
sexp	3342.80	2539.0	3167.17	(3296.54, 3389.07)
entropy	0.3983	0.0	0.8194	(0.39, 0.41)
days_to_first_fix	73.14	0.00	312.84	(68.58, 77.72)
exp	8293.86	7846.0	6150.00	(8204.03, 8383.71)

Table 11
Descriptive statistics of the model variables for Fabric 8

Variable	Mean	Median	SD	Confidence Interval
ndev	32.10	31.00	16.72	(31.82, 32.40)
ns	1.53	1.00	1.79	(1.51, 1.57)
nf	9.46	2.00	84.56	(7.98, 10.95)
rexp	186.125	1.20	3249.38	(129.14, 243.10)
sexp	895.08	179.00	1469.96	(869.31, 920.86)
entropy	1.0616	0.2108	1.5323	(1.03, 1.09)
days_to_first_fix	25.58	0.00	97.95	(23.86, 27.30)
exp	6918.40	3954.00	8384.15	(6771.39, 7065.43)

A.10. JGroup

The heat map of the correlation coefficients among the variables of interest in the JGroup data set is shown in Fig. 10.

Table 10 shows the statistics that can help to understand the distribution and variability of the different variables in the JGroup dataset.

A.11. Fabric 8

The heat map of the correlation coefficient among the variables of our interest for the fabric data set is shown in Fig. 11.

Table 11 shows the statistics that can help to understand the distribution and variability of the different variables in the fabric dataset.

A.12. Broadleaf

The heat map of the correlation coefficient among the variables of our interest for the Broadleaf data set is shown in Fig. 12.

Table 12 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table 12
Descriptive statistics of the model variables for Broadleaf

Variable	Mean	Median	SD	Confidence Interval
ndev	21.07	24.00	11.00	(20.88, 21.27)
ns	1.33	1.00	0.96	(1.32, 1.35)
nf	9.77	2.00	102.34	(7.97, 11.58)
rexp	129.34	1.09	879.91	(113.81, 144.86)
sexp	1643.81	327.50	3920.65	(1574.62, 1713.01)
entropy	1.0388	0.5586	1.3546	(1.01, 1.06)
days_to_first_fix	48.78	0.00	202.03	(45.21, 52.34)
exp	4628.11	1205.00	7577.76	(4494.37, 4761.85)

Table 13
Descriptive statistics of the model variables for Brackets

Variable	Mean	Median	SD	Confidence Interval
ndev	42.13	41.00	26.13	(41.66, 42.61)
ns	1.12	1.00	0.34	(1.11, 1.12)
nf	3.05	1.00	26.25	(2.58, 3.53)
rexp	43.76	1.61	264.21	(38.95, 48.57)
sexp	365.39	143.00	511.38	(356.09, 374.71)
days_to_first_fix	58.59	0.00	214.69	(54.69, 62.50)
entropy	0.4887	0.00	0.7824	(0.47, 0.50)
exp	668.61	246.00	1017.29	(650.10, 687.13)

A.13. Brackets

The heat map of the correlation coefficient among the variables of our interest for the Brackets data set is shown in Fig. 13.

Table 13 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Appendix

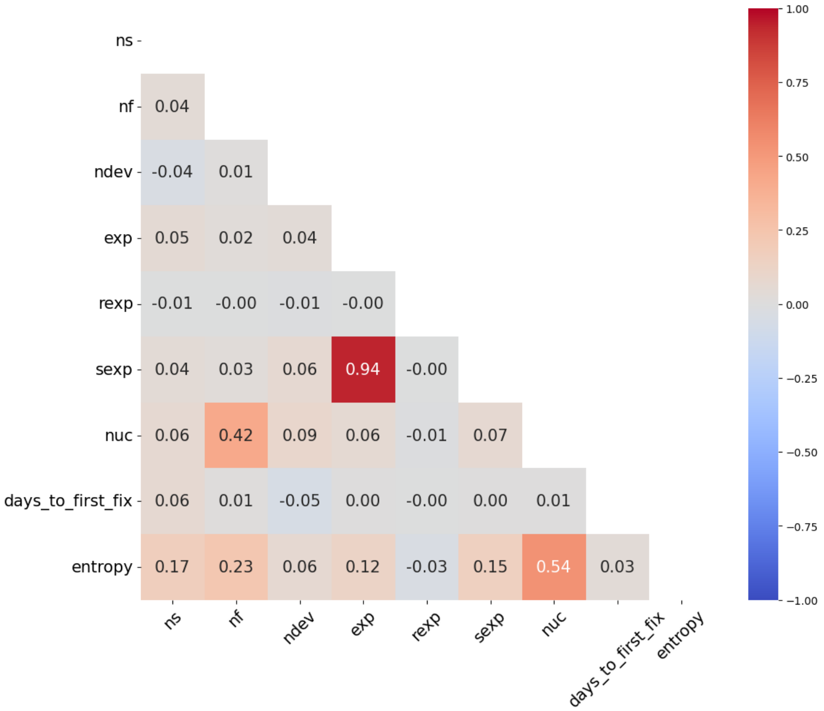

Figure 1: Heat map of correlation coefficients of Rust dataset

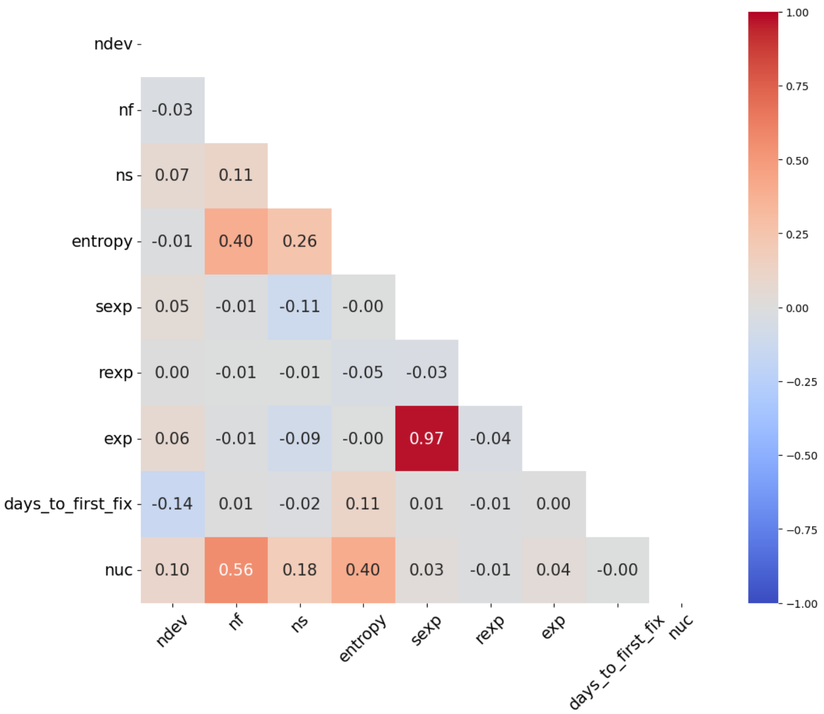

Figure 2: Heat map of correlation coefficients of Tensorflow dataset

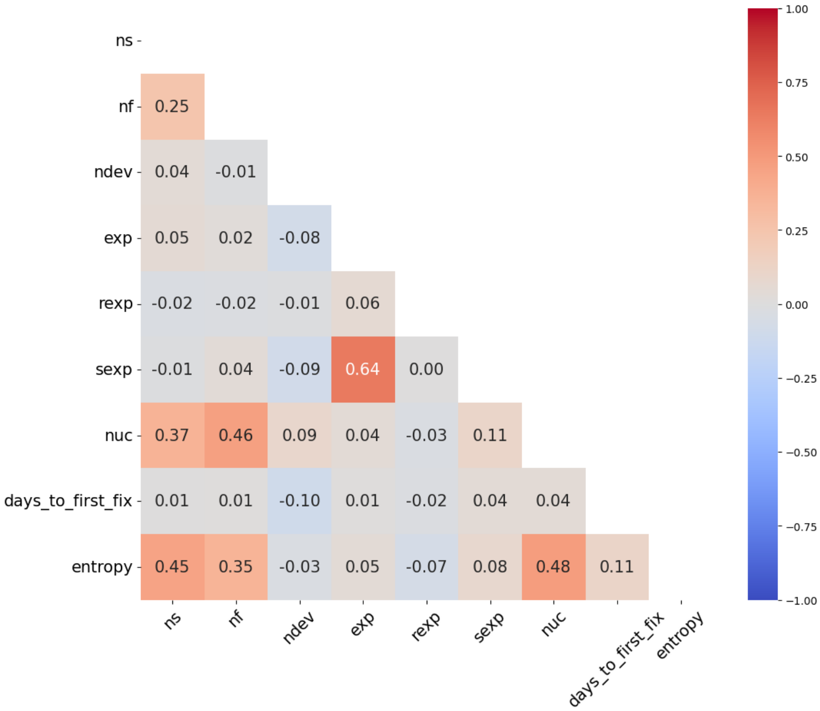

Figure 3: Heat map of correlation coefficients of Rails dataset

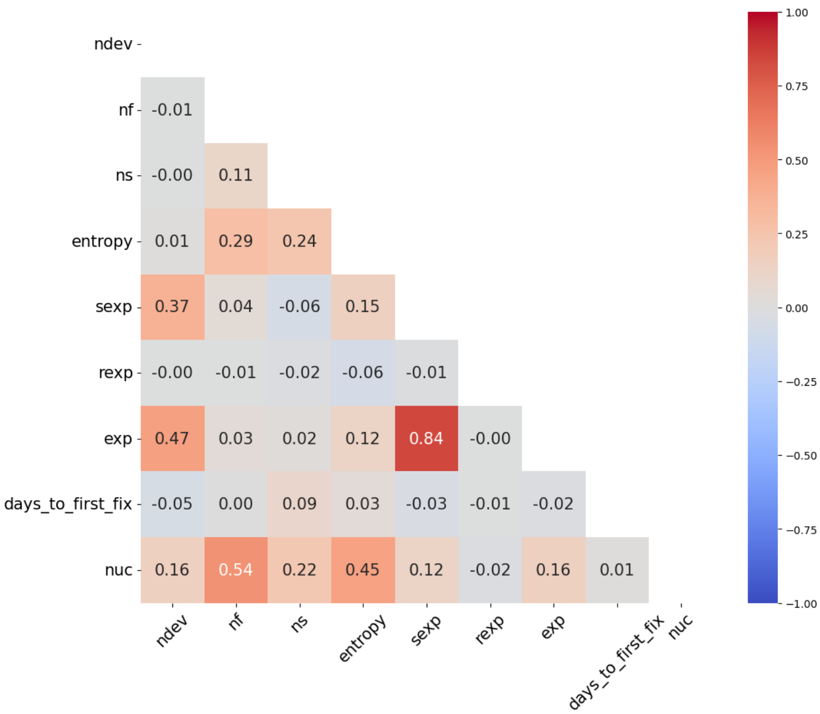

Figure 4: Heat map of correlation coefficients of VSCode dataset

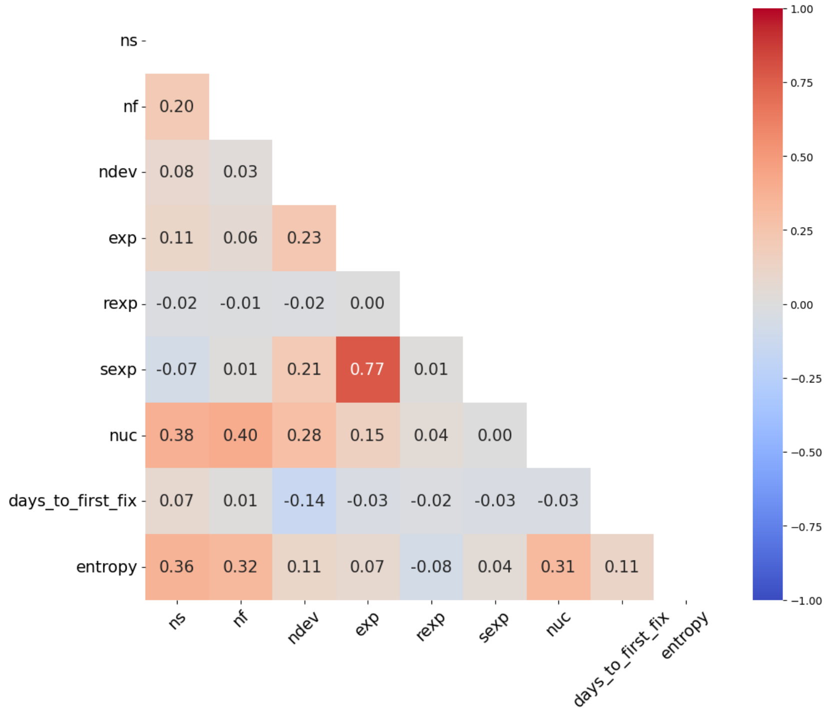

Figure 5: Heat map of correlation coefficients of WP-Calypso dataset

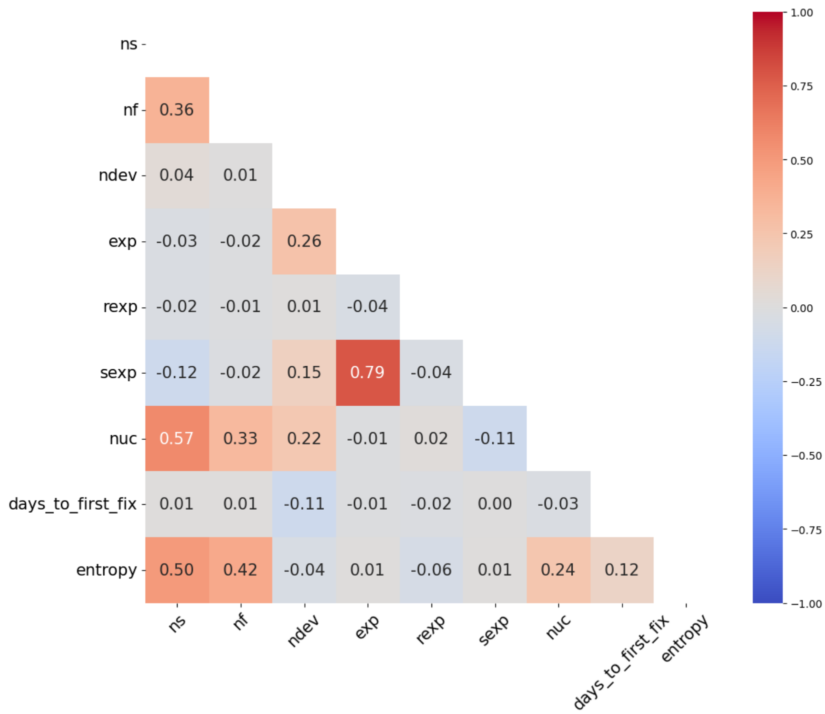

Figure 6: Heat map of correlation coefficients of Camel dataset

Appendix

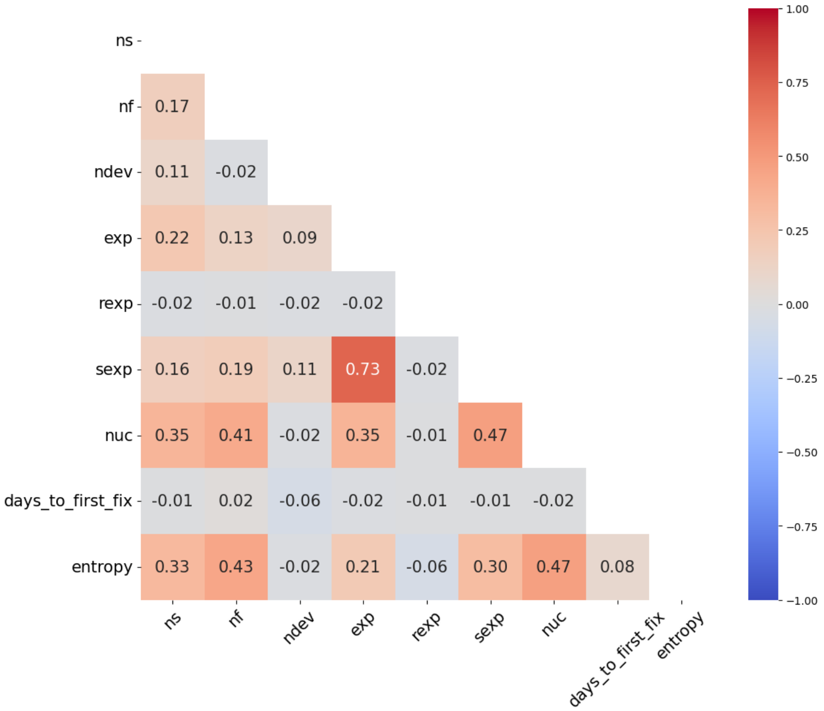

Figure 7: Heat map of correlation coefficients of CoreFX dataset

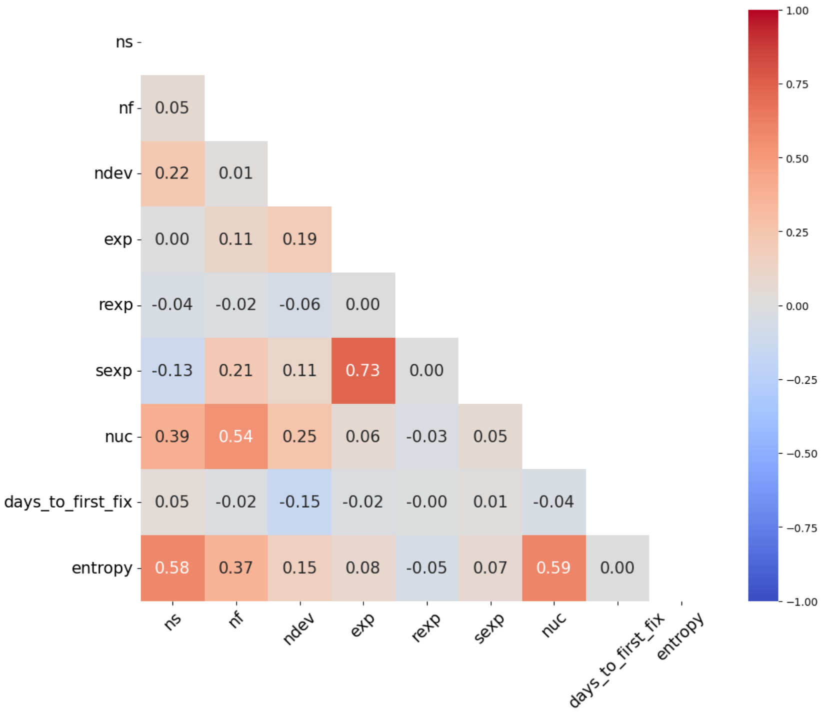

Figure 8: Heat map of correlation coefficients of Django dataset

Appendix

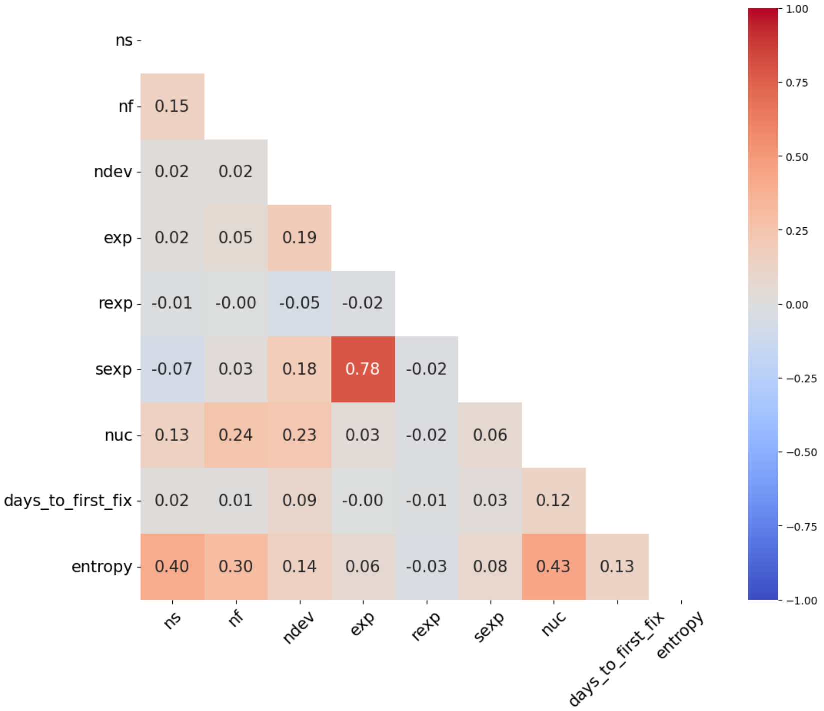

Figure 9: Heat map of correlation coefficients of Nova dataset

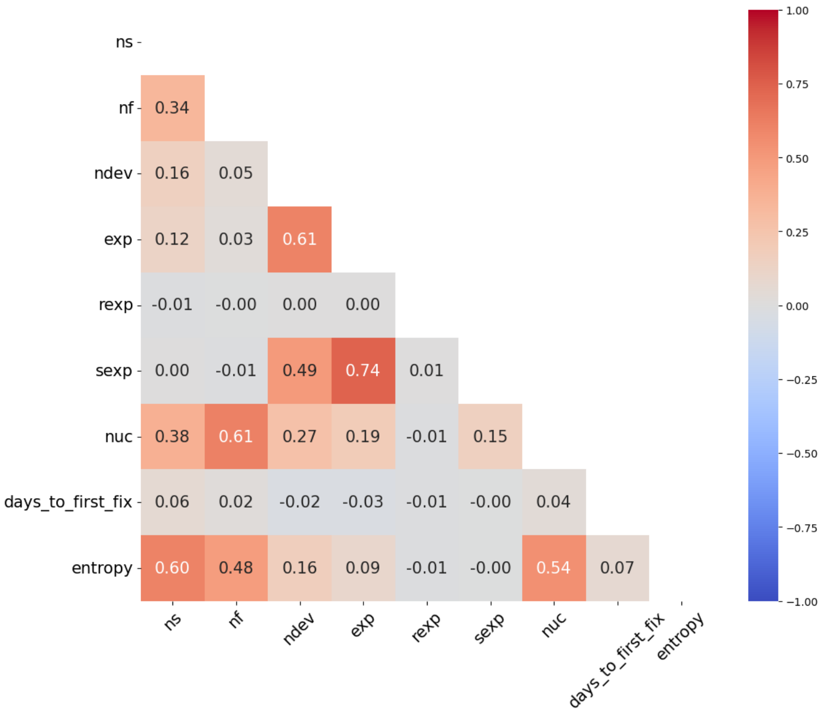

Figure 10: Heat map of correlation coefficients of JGroup dataset

Appendix

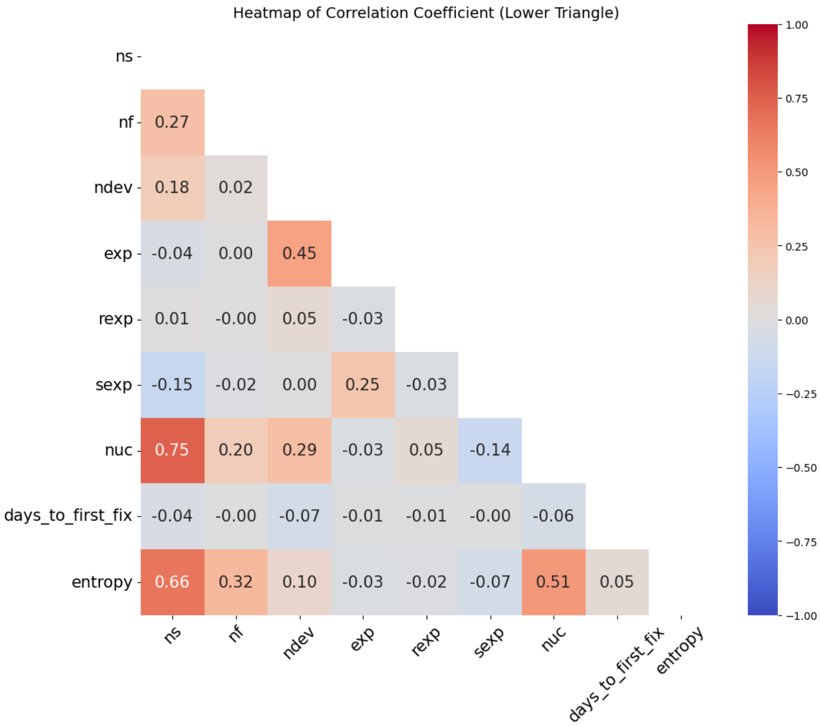

Figure 11: Heat map of correlation coefficients of Fabric 8 dataset

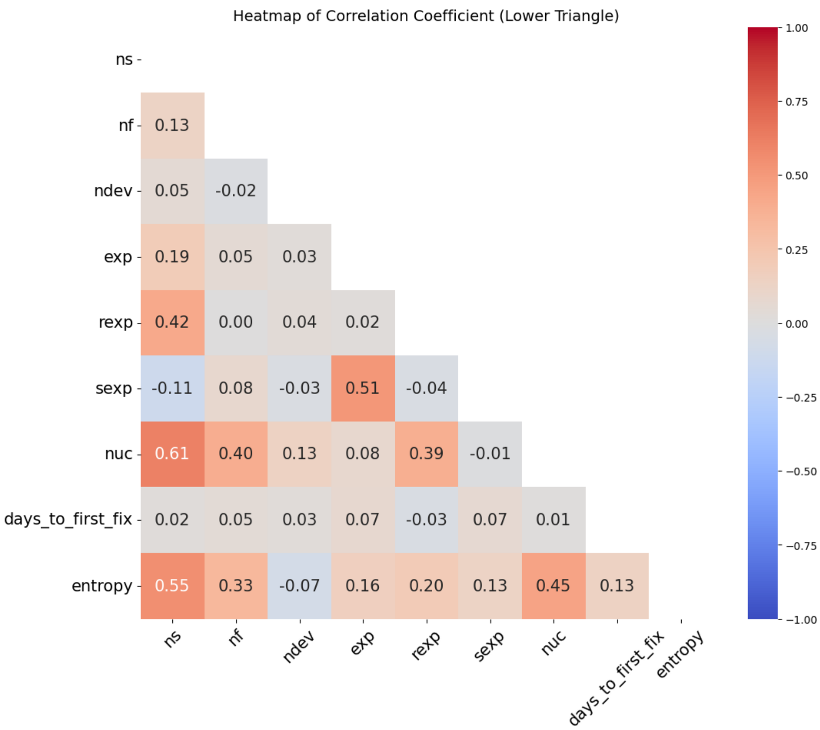

Figure 12: Heat map of correlation coefficients of Broadleaf dataset

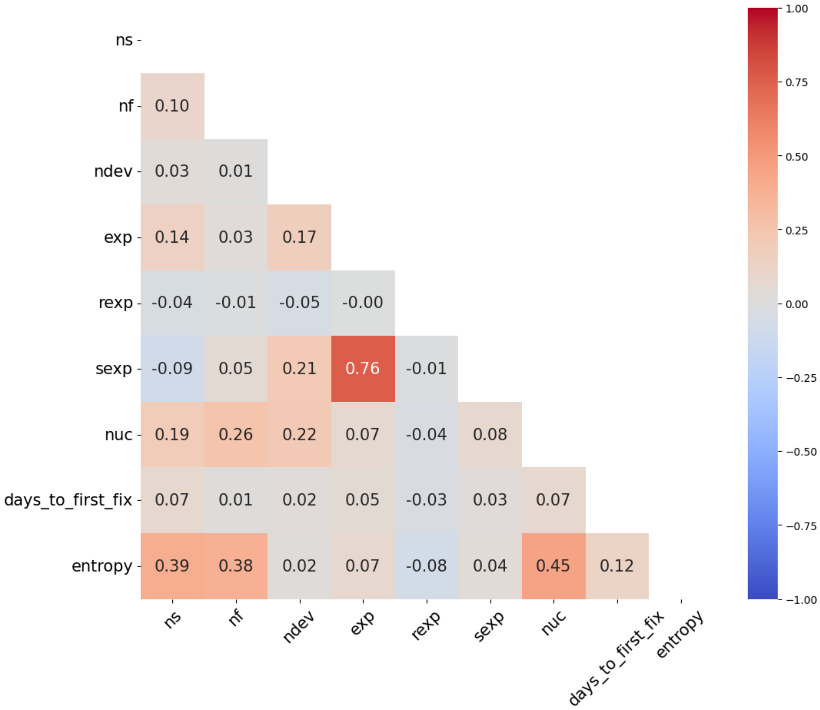

Figure 13: Heat map of correlation coefficients of Brackets dataset